

INFN Sezione di Bari & Dipartimento Interateneo di Fisica - Università di Bari
Medical Physics and Complex Systems Research Group

Deep Learning and Graph Metrics for Early Alzheimer's Diagnosis from Multi-Site MRI

Conferenza: IEEE International Symposium on Biomedical Imaging ISBI

Relatore: Prof.ssa Sabina Tangaro | Data: Aprile 2023

Punti Chiave della Presentazione / Slide Highlights:

- * Deep convolutional neural networks for automated feature extraction on 3D brain T1 scans.
- * Harmonization techniques for multi-site neuroimaging cohorts (ADNI, PPMI).
- * Cross-validation performance: $AUC > 0.92$ in differentiating MCI converters vs non-converters.
- * Clinical interpretability via saliency mapping and graph centrality indices.

Documento ad uso scientifico e didattico - <http://medphys.ba.infn.it/>